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Type	Article
Title	Microbial Bio-inoculation Effects on Seed Germination Dynamics and Field Performance of Pea (<i>Pisum sativum</i> L.) under Osmotic Stress and Fertilization
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Section	Sustainable Bioresource and Bioprocess Engineering
Abstract	Microbial bio-inoculants have been proposed as management tools to enhance crop performance under variable environmental conditions; however, their effectiveness is frequently modulated by site-specific factors. This study evaluated the effects of bio-inoculation on seed germination and seedling vigor of pea under osmotic stress induced by polyethylene glycol (PEG 6000), and assessed the interaction between bio-inoculation and two fertilization levels under field conditions in the Amazonas region of Peru. Under laboratory conditions, germination percentage remained high across all treatments (93.3–100%) and was not affected by bio-inoculation or osmotic potential. In contrast, osmotic stress significantly altered germination dynamics. Mean germination time increased from 1.85–2.09 days at 0 MPa to 2.26–2.43 days at –0.8 MPa, while germination synchrony and seedling vigor declined with increasing stress. The seedling vigor index reached maximum values at –0.2 MPa (4.47–5.29) and decreased at –0.8 MPa (1.50–2.00). Multivariate analyses indicated that variation in germination responses was mainly associated with germination timing and vigor rather than seed viability. Field experiments revealed variability in agronomic traits between locations. Plant height ranged from 38.5–46.3 cm in Lamud and from 100.6–108.3 cm in Molinopampa, while grain yield varied from 698–1846 kg/ha and 8771–9919 kg/ha, respectively. Yield variation was explained by pod weight and number of pods per plant. Environmental conditions showed a stronger influence than microbial bio-inoculation on germination dynamics and field productivity, although positive trends associated with bio-inoculant application were observed.

Review Report Form

Quality of English	(x) The English could be improved to more clearly express the research.
Language	() The English is fine and does not require any improvement.

	Yes	Can be improved	Must be improved	Not applicable
Does the introduction provide sufficient background and include all relevant references?	()	(x)	()	()
Is the research design appropriate?	()	()	()	(x)

Are the methods adequately described?	()	(x)	()	()
Are the results clearly presented?	()	()	(x)	()
Are the conclusions supported by the results?	()	(x)	()	()
Are all figures and tables clear and well-presented?	()	()	()	(x)

Comments and
Suggestions for
Authors

Review comments

This study focuses on pea (*Pisum sativum* L.) in the Amazonas region of Peru, investigating the effects of microbial bio-inoculation on seed germination dynamics and field performance through a combination of laboratory PEG-induced osmotic stress experiments and multi-location field fertilization trials. It also analyzes the interactive effects between bio-inoculation and different fertilization levels. The research design integrates laboratory mechanistic exploration and field agronomic validation, with standardized experimental methods and sufficient data support. The conclusions have certain reference value for the application of bio-inoculants in the sustainable production of peas in this region. However, the site selection is suboptimal, making it difficult to highlight the effects of microbial inoculation. The authors are required to address the following issues:

1. The field trials of this study were all conducted in the Amazonas region of Peru, and regional specificity is an important background of the research. Therefore, the study region can be added to the title. It is recommended to revise the title to: Microbial bio-inoculation effects on seed germination dynamics and field performance of pea (*Pisum sativum* L.) under osmotic stress and fertilization in **Peruvian Amazonas**. Please the authors carefully check and confirm.
2. The abstract only mentions "assessed the interaction between bio-inoculation and two fertilization levels under field conditions" but does not specify the specific parameters of the two fertilization levels (75% and 100% of the recommended dose), nor does it elaborate on the effects of fertilization levels alone or in synergy with bio-inoculation on the agronomic traits and yield of peas in the field. This prevents readers from obtaining key results of the core treatments of the study from the abstract, weakens the completeness of the study's experimental design and the pertinence of the results, and is disconnected from the

actual field experimental design and result analysis of the study. Please supplement this content. At the same time, a brief mention of the core significance/application value of the study can be added at the end of the abstract to enhance the completeness of the abstract and highlight the practical value of the study. Meanwhile, Keywords need to be revised, supplemented, and sorted according to the core content of the study (fertilization, microbial bio-inoculation, pea, osmotic stress/drought stress) to make them more in line with the research theme and journal retrieval standards. Please revise the keywords again.

3. In the result analysis, Tables 1 and 2 should adopt the three-line table format, and units do not need to be listed for each entry. For example, in Table 1, "(CFU/mL or CFU/g)" can be added after "Concentration"; in Table 2, the corresponding units can be included in the "Parameter" column. It is recommended to remove the background grid lines in all the Figures.
4. When discussing that "the low-fertility, alkaline soil at the Lamud site leads to the failure of microbial colonization, thereby limiting the effect of bio-inoculation", the study only infers indirectly through "absence/scarcity of nodulation" and lacks direct determination data of microbial colonization (such as the population size of inoculated microorganisms in the rhizosphere, colonization rate, etc.). This results in insufficient support for this key conclusion and an incomplete logical chain. It is recommended to use the **plate counting method** (suitable for culturable microorganisms) or **qPCR quantitative analysis** (targeted detection of characteristic microorganisms in the inoculants, such as *Azospirillum brasilense* in Rizoplant and *Bacillus subtilis* in Amysub) to determine the population density of inoculated microorganisms in the rhizosphere soil (CFU/g soil or gene copies/g soil), so as to further verify your hypothesis.
5. The results of the laboratory germination experiment and field experiment are independent of each other, and the connection between them is not clarified, leading to an insufficiently tight cross-scale connection between "laboratory mechanism and field agronomy". It is recommended to add a "correlation analysis between germination indicators and field traits": select core laboratory indicators (such as seedling vigor index, SVI; mean germination time, MGT) and key field traits (plant height, pod weight, yield) for Pearson or Spearman correlation analysis to clarify whether early germination performance can predict field productivity.

6. The study only presents the comparison results of 75% and 100% fertilization doses, but does not clarify the effect of no fertilization, making it impossible to fully highlight the compensatory effect of "bio-inoculation + reduced fertilization". It is recommended to add data with no fertilization as a control.
7. The soil physical and chemical properties of Lamud (alkaline, low phosphorus, low organic matter) and Molinopampa (acidic, high phosphorus, high organic matter) are extremely different, and both have problems such as extreme pH and unbalanced nutrient supply. This is the core reason why microbial inoculants are difficult to colonize and their effects are weak—extreme soil conditions either inhibit the metabolic activity of inoculated microorganisms (such as the alkaline environment of Lamud) or leave no room for the growth-promoting functions of microorganisms (such as the high nutrient supply of Molinopampa). Ultimately, the study cannot accurately evaluate the true potential of microbial inoculants. Therefore, it is recommended that the authors replace the actual field sites and select sites with more suitable soil conditions for pea growth and microbial colonization for detailed analysis. At suitable sites, intermediate processes such as microbial colonization efficiency, pea nodulation, and nutrient absorption can be observed more clearly, providing direct evidence for the "compensatory effect of bio-inoculation" and deepening the mechanistic level of the study. The selection of the two sites in this paper weakens the role of microbial inoculation, making it difficult to analyze its impact on pea seed germination and agronomic traits.

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